The Journal of Open Source Software

pangeo-fish: A Python package for studying fish movement using biologging and earth science data

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DOI: 10.xxxxxx/draft

Software

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Editor:

Submitted: 07 March 2025
Published: unpublished

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Summary

Geo-referenced data plays an important role in understanding and conserving natural resources, particularly when investigating biological phenomena such as fish migration and habitat uses. Biologging, the practice of attaching small devices (called *tags*) to animals for recording behavior, physiology, and environmental data, proves to be invaluable in this field.

As fish can not be tracked directly using tracking devices such as GPS receivers, geolocation models have emerged to estimate fish positions by correlating individual series of physical measurements — e.g. temperature and pressure records — with geophysical reference fields - oceanic temperature and bathymetry - derived from satellite observations and hydrodynamical model outputs. The quality of geophysical reference fields is therefore of paramount importance, which depends on the spatio-temporal resolution at which the physical processes are considered and modeled. However, higher resolutions involves more data, that requires significant computing power and storage capacity.

To address this challenge, we developed a Python package for fish tracking estimation, named **pangeo-fish**. It is based on the Pangeo ecosystem, which offers a unique interoperable, scalable, open source tools for interactive data analysis in the field of big data marine and geoscience.

pangeo-fish is dedicated to ecologists to process recovered archival tag data and fish tracking estimation. These fish geolocation models are critical for better understanding fish movements and behaviors, and are nowadays seen as a powerful tool by policy makers to improve fish management and conservation.

Statement of need

Bio-logging consists in attaching onto (or sometimes inserting into) an animal an electronic device that will record in its memory physical and/or geochemical parameters as a function of time so that scientists can reconstruct the activity of the animal, the characteristics of the environment it travels in and the interactions between the two. These tools can provide a wealth of information on the behaviors and movements of free-swimming marine animals, including diving and activity patterns, habitat selection, swimming behavior, mating behavior, energy use, interaction with environment. However, unlike terrestrial animals, marine mammals or large sharks (which regularly return to the surface), whose positions can be directly estimated using ARGOS or GPS technologies, tracking fish underwater is challenging.

Figure 1: Promotion of the FISH-INTEL tagging campaign.

For studying fish movements, the two widely used electronic tagging technologies are acoustic telemetry and archival tags. Acoustic telemetry involves a tag that emits an acoustic signal containing a unique ID and possibly sensor data. This signal can be detected by an acoustic receiver when the tagged animal is within range, and the detection data is retrieved from the receiver. As such, acoustic tags do not need to be recovered, but there is no guarantee that the tagged fish will swim around the receivers network.

In contrast, archival tags store sensor measurements at set intervals in their memory. To access the logged data, these tags must either be recovered (which mostly depends on fishers and the local population living along the coast) or transmit their information via satellite. The data from archival tags can offer detailed insights into vertical movement patterns (Heerah et al., 2017), environmental preferences (Carla et al., 2021; Righton et al., 2010), and can be used to reconstruct migration paths through geolocation modeling.

Over the past two decades, several geolocation methods have been developed to track fish movements. The first approach compares tag records with field observations of variables like light intensity and temperature, assuming fixed swimming speeds [Metcalfe & Arnold (1997); hunter2003geolocation]. A more advanced method uses state-space models (SSMs), which are flexible statistical tools that estimate the likelihood of unknown states, such as fish location and behavior (Patterson et al., 2017; Royer et al., 2005). SSMs can account for uncertainties in both tag records and field observations.

Hidden Markov Models (HMMs) are a specific type of SSM which can be adapted to be suitable for geolocation. The resulting class of HMMs represents fish locations as discrete states within a spatial grid and consist of two sub-models: the process model, which predicts movement between grid cells, and the observation model, which predicts the likelihood of observations based on the state. The estimation process involves iterative "time updates" and "data updates" to refine the fish's location over time, culminating in a "forward filter" and a "backward filter" to incorporate all available data (Pedersen et al., 2008; Woillez et al., 2016).

Figure 2: Example of an acoustic tag (on the left) and a DST (on the right). See the centimeter scale for size reference.

In practice, estimating fish positions relies on the likelihood of observed data from archival tag logs, such as temperature at specific depths, combined with reference geoscience data like satellite observations and oceanic temperature from hydrodynamic models. The quality of these geophysical reference fields is crucial and depends on the spatio-temporal resolution at which physical processes are modeled. A first approach to enhance the accuracy of geolocation model predictions involved using additional information. Acoustic tags provide location data when tagged animals are within range of a receiver, while archival tags offer continuous time series of logged sensor measurements, from which trajectories can be inferred. These technologies provide complementary information on fish movement patterns, helping to overcome the limitations of each method. Fish trajectories can be reconstructed from logged depth and temperature histories using an existing geolocation modeling approach, adapted to include the likelihood of the acoustic detections (Gonse, 2024; Goossens et al., 2023). A second approach involved the use of more accurate hydrodynamic models with high spatial and temporal resolutions. However, higher resolutions involves more data, that requires significant computing power and storage capacity.

The Pangeo community handles these challenges, by fostering an ecosystem of interoperable, scalable, open source tools for interactive data analysis in the field of big data marine and geoscience. Therefore, the Pangeo ecosystem represents a powerful mean through which ecologists can analyze more easily their archival tag data and improve the accuracy of geolocation model predictions, which will help to better understand fish ecology, essential fish habitats, migration pathways, all the information needed for the sustainable management of a fishery resource (de Pontual et al., 2019, 2023).

Unfortunately, the research community lacks of adaptable, scale and open source implementations of geolocation models. **pangeo-fish** is a Python package that fills this gap.

As its name suggest, the software has been designed to be used within the Pangeo ecosystem on several aspects, therefore accounting for both the users' needs (user-friendly API and meaningful result visualization) and the computational challenges. In particular, **pangeo-fish** has a robust data model based on Xarray and scales computation with Dask.

Data loading processes are furthermore streamlined by libraries like `intake`, `kerchunk` or `fsspec`, and the previously mentioned `xarray` data model enables interactive visualization of the results thanks to tools such as the `hvplot` library and the JupyterLab environment.

Similarly, `pangeo-fish`'s I/O operations are automatically distributed with the combination of Dask and Zarr. The Pangeo software stack provides researchers with the necessary tools to access reference data and perform intensive computations in a scalable and interactive manner. `pangeo-fish` gives ecologists a user-friendly tool for inferring fish locations from archival tag data, hence filling the gap between their expertise and the Pangeo's environment capabilities.

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Geolocation Model

pangeo-fish implements a method well established in the fish trajectory reconstruction literature Goossens et al. (2023). It consists of a Hidden Markov Model introduced above.

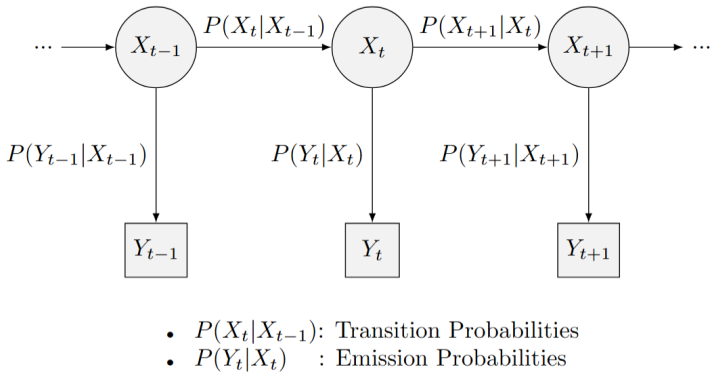

Figure 3: Illustration of the Hidden Markov Model. The hidden states Xt describe the fish's positions, and the emission probabilities $P(Y_t|X_t)$ correspond to the likelihood of observing the fish at time t.

As illustrated in Figure 3, the latent (or *hidden*) states X_t of the HMM infer the (daily or hourly) fish's positions, and the observation process relates the sensor records with the environmental reference data. The transition matrix between the hidden states is modeled as a Brownian motion parametrized by the standard deviation σ. As such, fitting the geolocation model for a tag's records aims to both estimate the value of σ that maximizes the likelihood of the hidden states sequence (i.e., the fish's trajectory) given the observations, and a reconstruction of the sequence of hidden states from observed positions, e.g. the most probable track (the Viterbi algorithm), the mean and the modal tracks. The optimal likelihood value reflects the level of residual inconsistency between the observed (recorded) tag data and the reference data.

Key Features of pangeo-fish

pangeo-fish is a Python software that handles the entire pipeline for reconstructing fish trajectories given sensor records and a reference data, and visualizing the results. The key features of **pangeo-fish** are its simple API that follows each stage of the pipeline, and its scalability, which includes parallel computation and streamlined remote data fetching.

Pre-processing

The framework starts by loading the archival records of a tagged fish:

```
from pangeo_fish.helpers import load_tag

tag, tag_log, time_slice = load_tag(tag_root, tag_name, storage_options)
```

Then, the user selects the reference model (depending on, for example, the studied area or the resolution of the model):

```
from pangeo_fish.helpers import load_model

reference_model = load_model(uri, tag_log, time_slice, ...)
```

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In the previous instruction, the oceanic model is reduced to fit the scope of the tag (in time and space) `load_model()`.

The next step involves complex operations, which are however made easily accessible for the user in a few instructions. The operations consists of computing the difference between the archival tag and reference data, reshaping the consequent result (to avoid spatial distortions) to finally compute the emission probabilities given the initial (and optionally the final) position(s) of the fish and the reshaped dataset:

```
from pangeo_fish.helpers import compute_diff, regrid_dataset, compute_emission_pdf

diff = compute_diff(reference_model, tag_log, relative_depth_threshold, ...)[0]
reshaped = regrid_dataset(diff, nside, min_vertices, ...)[0]
emission_pdf = compute_emission_pdf(differences, tag["tagging_events"].ds, ...)[0]
```

Accounting for acoustic detections

Before normalizing the emission probabilities, the user can include another distribution, based on the possible fish detections by acoustic receivers:

```
from pangeo_fish.helpers import compute_acoustic_pdf, combine_pdfs

acoustic_pdf = compute_acoustic_pdf(emission_pdf, tag, receiver_buffer, ...)
combined_pdf = combine_pdfs([emission_pdf, acoustic_pdf], ...)
```

Model optimization

Finally, the normalized distributions are fitted to find the optimal σ, and the fish's positions can be extracted (**pangeo-fish** features several options for reconstructing trajectories):

```
from pangeo_fish.helpers import optimize_pdf, predict_positions

parameters = optimize_pdf(combined_pdf, maximum_speed, save_parameters=True, ...)
states, trajectories = predict_positions(path_to_previous_results, track_modes, ...)
```

Result analysis and visualization

Every result of the step described above can be easily visualized for analysis with **pangeo-fish**. For instance, the time series of the archival tag's can be easily visualized with:

```
from pangeo_fish.helpers import plot_tag

plot = plot_tag(tag, tag_log, save_html=True, ...)
```

The user can also plot an interactive visualization of the trajectories:

```
from pangeo_fish.helpers import plot_trajectories

plot = plot_trajectories(path_to_previous_results, track_modes, save_html=True)
```

As for the state and emission distributions, a video of their evolution (alongside of each other) can be rendered:

```
from pangeo_fish.helpers import open_distributions, render_distributions

data = open_distributions(path_to_previous_results, storage_options, ...)
render_distributions(data, "results/", extension="mp4", remove_frames=True, ...)
```

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Conclusion

pangeo-fish is a Python package that implements a geolocation model, based on a Hidden Markov Model, for estimating fish positions from archival tag and oceanic data. Designed to work with the Pangeo ecosystem, it aims to support the ecologists with their research, by handling backend processes — such as data loading or parallel computation — while exposing a user-friendly interface to manage their archival tag data and run the geolocation model.

Acknowledgements

- T Odaka, JM Delouis and J Magin are supported by the CNES Appel, a projet R&T R-S23/DU-0002-025-01.
- T Odaka, JM Delouis and M Woillez are supported by the TAOS project funded by the IFREMER via the AMII OCEAN 2100 programme.
- Q Mazouni, M Woillez, A Fouilloux and T Odaka are supported by Global Fish Tracking System (GFTS), a Destination Earth use case procured and funded by ESA (SA Contract No. 4000140320/23/I-NS, DESTINATION EARTH USE CASES - DESP USE CASES - ROUND (YEAR 2023))
- M Woillez and T Odaka are supported by Digital Twin of the Ocean - Animal Tracking (DTO Track), a European project that aims to create a digital twin of the North Sea using animal tracking data and funded by the Sustainable Blue Economy Partnership (ANR-24-SBEP-0001)

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